Martin Mathew

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SUMMARY

Computational Engineering graduate with an M.Sc. from TU Dresden, specializing in CFD, computational fluid mechanics, numerical modelling, and scientific computing. Experienced in developing and analyzing numerical simulations for hydrodynamics, multiphysics, turbulence, multiphase flows, and complex engineering systems using STAR-CCM+, ANSYS Fluent, OpenFOAM, Python, MATLAB, and Linux/HPC environments. Research experience includes CFD-based hydrofoil control, multiphysics simulations at Fraunhofer IKTS, and collaborative research with Chalmers University of Technology. Strong interest in advanced computational methods, data-driven modelling, scientific machine learning, and simulation-driven engineering. Motivated to contribute to interdisciplinary research and industrially relevant engineering projects through rigorous numerical analysis, problem solving, and continuous methodological development.

TECHNICAL SKILLS

- **CFD & Simulation Tools:** STAR-CCM+, ANSYS Fluent, ANSYS Rocky, ANSYS ICEM CFD, OpenFOAM
- **Methods:** RANS turbulence modelling, LES (exposure), multiphase modelling, mesh sensitivity studies, transient simulation, simulation-test correlation and validation
- **Programming & Automation:** Python (simulation automation, post-processing), MATLAB/Simulink, Java
- **Post-Processing:** ParaView, CFD-Post, EnSight
- **Other:** LaTeX, closed-loop control integration (PID), HPC/Linux-based computational workflows
- **Languages:** English (C1), German (A2)
- **Core Strengths:** Cross-functional collaboration, structured analytical problem-solving, technical documentation and reporting, adaptability across multicultural teams
- **Research Practice:** Solver validation against reference/experimental data, reproducible workflow design, benchmark case development, technical documentation and reporting

PRACTICAL EXPERIENCE

Research Intern | University of Split, Croatia | Mar 2026 – Aug 2026

- Validated CFD solver accuracy across a set of 2D benchmark flow cases against analytical and reference datasets, establishing a reproducible validation baseline for extension to 3D.
- Investigated the influence of mesh resolution, time-stepping, and solver settings on solution quality to identify robust, reproducible simulation parameters.
- Built a validated 2D case framework designed for extension to 3D benchmark studies intended to run on HPC systems.
- Developed reproducible workflows for simulation setup and post-processing to support consistent results across studies.

Student Research Assistant | Fraunhofer IKTS, Dresden, Germany | May 2024 – Sep 2025

- Automated multi-case CFD workflows in Python, running batch parameter studies (e.g., pressure drop across a range of inlet velocities) from a single script to eliminate manual case setup and execution.
- Performed CFD studies on particle bed reactors using ANSYS Fluent, generating and refining high-quality meshes in ANSYS ICEM CFD supported by systematic mesh sensitivity studies.
- Collaborated with experimental teams to validate simulation results against lab-scale measurements, supporting simulation-to-physical-test correlation.
- Delivered automated data analysis and reporting tools used by internal stakeholders to accelerate results review.

M.Sc. Thesis Exchange Semester | Chalmers University of Technology, Sweden | Sep 2024 – Mar 2025

- Designed and built a coupled CFD-control simulation framework (STAR-CCM+, Java automation, MATLAB/Simulink) for hydrofoil hydrodynamics, engineered for reuse so future researchers could develop and test more sophisticated controllers.
- Implemented closed-loop PID control to regulate hydrodynamic objectives within the CFD environment.
- Integrated Java-based automation to couple control logic directly with the CFD solver, removing manual intervention between control and simulation steps.
- Pre-validated control strategies in MATLAB/Simulink across a range of operating conditions prior to CFD integration.

EDUCATION

M.Sc. Computational Engineering | TU Dresden, Germany | Oct 2022 – Sep 2025

GPA 2.2 (good). Built strong foundation in discretization techniques, numerical stability, and numerical accuracy relevant to industrial CAE workflows.

B.Tech. Mechanical Engineering | APJ Abdul Kalam Technological University, India | Aug 2016 – Oct 2020

8.41/10 (very good). Coursework: Mechanics of Solids, Mechanics of Fluids, Thermodynamics, Design of Machines, Automotive Engineering, Fluid Machinery.

AWARDS & SCHOLARSHIPS

- DAAD Scholarship (Erasmus+), Exchange Semester, Chalmers University, Gothenburg, Sweden — 2024
- DAAD Scholarship (Erasmus+), Exchange Traineeship, Žilina, Slovakia — 2025

INTERNATIONAL EXPERIENCE

Worked across Germany, Sweden, Croatia, Slovakia, and India in multicultural, interdisciplinary research environments.

RESEARCH INTERESTS

- Large Eddy Simulation (LES)
- Marine Hydrodynamics
- Particle Simulation
- Hydraulic Machinery
- Cavitation
- Multiphase Flow
- FSI
- Multi- physics Modelling
- CFD + Control System Integration
- High Performance Computing